

Exam II : Analysis 2.

(1H45)

Exercise 1. (8 pts)

1. Solve the following equation

$$y''(x) + my(x) = 0, \quad \forall x \in \mathbb{R} \quad (E),$$

where m is an arbitrary real number.

2. Integrate the following equation

$$y''(x) - y(x) = (4x+6)e^x, \quad \forall x \in \mathbb{R} \quad (F).$$

3. Find the general solution of the following equation

$$y''(x) + y(x) = \sin^2 x \quad \forall x \in \mathbb{R} \quad (G).$$

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Exercise 2. (7 pts) Let the function f be defined on $\mathbb{R}^2$ by

$$f(x,y) = \begin{cases} \frac{x^3}{x^2 + (y-1)^2} & \text{if } (x,y) \neq (0,1) \\ 0 & \text{if } (x,y) = (0,1). \end{cases}$$

1. Study the continuity of f on $\mathbb{R}^2$.
2. Calculate the gradient of f , $\nabla f(x,y)$, for all $(x,y) \in \mathbb{R}^2$.
3. Is the function f of class C^1 on $\mathbb{R}^2$?
4. Is the function f differentiable on $\mathbb{R}^2$?

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Exercise 3. (5 pts) Let the function f be defined on $\mathbb{R}^2$ by

$$f(x,y) = xy - \frac{1}{3}x^3 - \frac{1}{3}y^3 + 4.$$

- (I) Give the Taylor formula of the function f of order 2 in the neighborhood of $(1,2)$
- (II) Study the extrema of f .